02/02/2026

Toric geometry V:

Properness & polytopes

Last time Functoriality, Smoothness

Today Properness, polytopes, recollections on divisors

> Last time we showed:

Proposition Let N be a lattice of rank n and $\sigma \subset N_{\mathbb{R}}$ a cone of dimension d. Then the following are equivalent:

(1) The scheme X_σ is smooth over $\mathrm{Spec}(\mathbb{Z})$ (hence the same is true after every basechange)

(2) There are generators for σ (as a cone) that can be extended to a basis for the lattice N

(3) There exists an isomorphism

$$X_\sigma \cong \mathbb{A}^d_{\mathbb{Z}} \times (\mathbb{G}_m)^{\times(n-d)}$$

Properness

Def. Let (N, Σ) be a lattice and a fan. The **support** of Σ is

$$|\Sigma| := \bigcup_{\sigma \in \Sigma} \sigma \subset N$$

Proposition.

(1) For a fan (N, Σ), the toric variety X_Σ is proper if and only if $|\Sigma| = N$.

(2) For a map of fans $\phi : (N, \Sigma) \longrightarrow (N', \Sigma')$, the induced morphism

$$\phi_* : X_\Sigma \longrightarrow X_{\Sigma'}$$

is proper if and only if $\phi_{\mathbb{R}}^{-1}(|\Delta'|) = |\Delta|$

Def. A fan (N, Σ) is **proper** if $|\Sigma| = N$.

Remark. Because of classical terminology from varieties, proper fans are also called **complete**

Remark. The proof can be found in §2.4 of Fulton's book. Technically, he works over $\mathbb{C}$, but the proof can be remidied as follows

(1) The proof that $\phi_{\mathbb{R}}^{-1}(|\Delta'|) = |\Delta|$ implies properness just uses the valuative criterion for properness. This works verbatim the same over $\mathbb{Z}$

(2) For the converse, note that since properness is stable under base change,

$$\phi_*: X_\Sigma \to X_{\Sigma'} \text{ proper} \Rightarrow X_{\Sigma,\mathbb{C}} \longrightarrow X_{\Sigma',\mathbb{C}}$$

is proper. Fulton then uses classical methods to show that if $\phi_{\mathbb{R}}^{-1}(|\Delta'|) \neq |\Delta|$, then $X_{\Sigma,\mathbb{C}} \to X_{\Sigma',\mathbb{C}}$ is not proper. So with (1) this is sufficient for what we want.

> Presumably one can directly show that if $\phi_{\mathbb{R}}^{-1}(|\Delta'|) \neq |\Delta|$, then ϕ_* does not satisfy the valuative criterion for properness. This would be better.

Background on polytopes

Def Let V be a fd $\mathbb{R}$-vector space. A subset $K \subset V$ is a **convex polytope** if K is the convex hull of finitely many points.

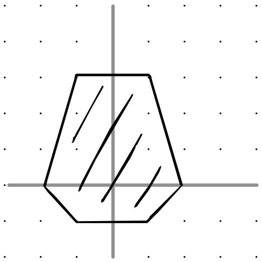
> A **proper face** F of a convex polytope K is the intersection with a supporting affine hyperplane

$$F = K \cap \{v \mid \langle u, v \rangle = r\}$$

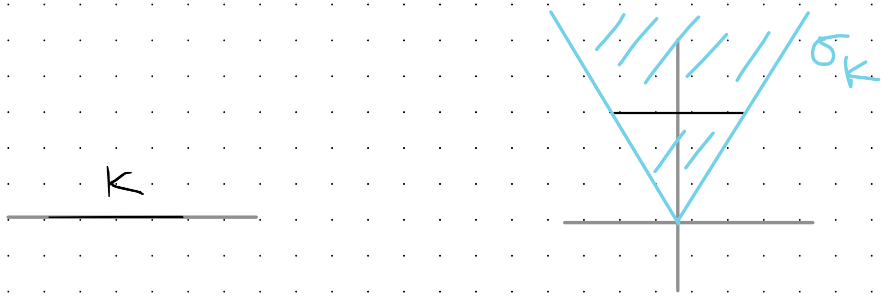
where $r \in \mathbb{R}$ and $u \in V^\vee$ is such that $\langle u, v \rangle \geq r$ for $v \in K$.

> We'll also regard $\emptyset$ and K as faces

Ex



Observe If $K \subset V$ is a convex polytope, we can form the cone σ_K on $K \times \{1\}$ in $V \times \mathbb{R}$



Then there is an order-pesenving bijection

$$\mathrm{Face}(K) \xrightarrow{\sim} \mathrm{Face}(\sigma_K)$$

$$F \longmapsto \sigma_F$$

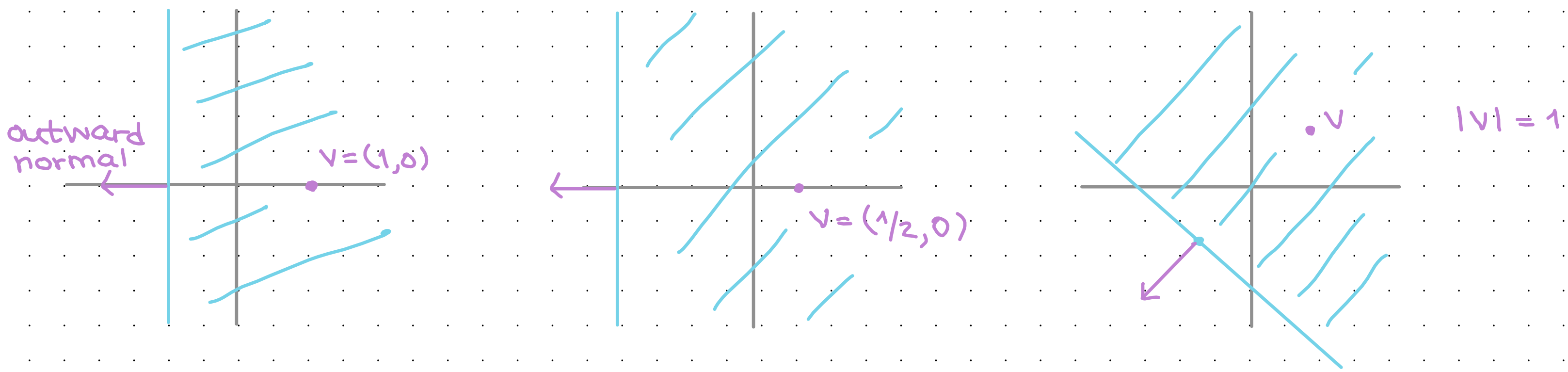
$$\varnothing \longmapsto 0 \quad \text{origin}$$

This can be used to deduce all of the standard properties of faces we discussed for cones

Def. If $K \subset V$ is a convex polytope, the **polar** of K is

$$K^\circ := \{ u \in V^\vee \mid \text{for all } v \in K, \text{ we have } \langle u, v \rangle \geq -1 \}.$$

Ex Let $K = \{v\}$ consist of a single nonzero element. Then K° is the half-space whose boundary hyperplane's outward normal is the direction $-v$ with distance from the origin $\frac{1}{|v|}$.



So if we write $H_v = \{v\}^\circ$ for this half space, then

$$K^\circ = \bigcap_{v \in K} H_v.$$

Ex.

The polar of a cube with vertices $(\pm 1, \pm 1, \pm 1)$ is the octahedron with vertices $\pm e_i^*$.

Duality for convex polytopes. Let $K \subset V$ be a convex polytope of dimension $\dim(V)$ containing the origin in its interior. Then

(1) K° is a convex polytope

(2) $(K^\circ)^\circ = K$

(3) For a face $F \subset K$, write

$$F^* := \{u \in K^\circ \mid \text{for all } v \in F, \text{ we have } \langle u, v \rangle = -1\}.$$

Then F^* is a face of K° and

$$\dim(F) + \dim(F^*) = \dim(V) - 1$$

(4) The map of posets $(-)^* : \mathrm{Face}(K)^{op} \longrightarrow \mathrm{Face}(K^\circ)$ is an isomorphism

(5) If $V = N_{\mathbb{R}}$ and K is rational, then $K^\circ \subset M_{\mathbb{R}}$ is also rational.

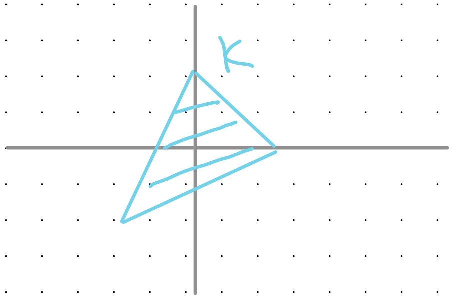
Fans from polytopes

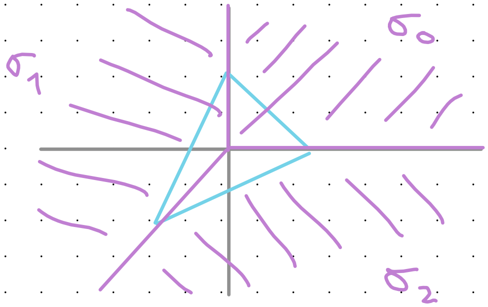
Example Let $K \subset N_{\mathbb{R}}$ be a rational convex polytope of dim rank(N) that contains the origin in the interior. Then the cones

$$C_K := \{ \text{cone over } F \mid F \subset K \text{ proper face} \}$$

form a fan in N.

> For example, let K be the convex hull of $e_1, e_2, -(e_1+e_2)$





The resulting fan is the fan we used to construct $\mathbb{P}^2$

Construction. Let $P \subset M_{\mathbb{R}}$ be a rational polytope of dimension rank(M). For each proper face $Q \subset P$, write

$$\sigma_Q^\vee := \text{Cone}\{u' - u \mid u' \in P, u \in Q \text{ vertices}\}$$

and $\sigma_Q = (\sigma_Q^\vee)^\vee \subset N_{\mathbb{R}}$. The **normal fan** associated to P is the fan in N given by

$$\Sigma_P = \{\sigma_Q \mid Q \subset P \text{ is a nonempty face}\}$$

Note that $\sigma_P^\vee = M_{\mathbb{R}}$, so $\sigma_P = 0$

> Unwinding definitions, one can show that

$$\sigma_Q = \{v \in N_{\mathbb{R}} \mid \text{for all } u \in Q \text{ and } u' \in P, \langle u, v \rangle \leq \langle u', v \rangle\}$$

- σ_Q is dual to the "angle" at Q consisting of all vectors pointing from points of Q to points of P

Note. The maximal cones of Σ_P are σ_v for $v \in P$ a vertex. So one can also just define Σ_P as the fan whose maximal faces are the σ_v

Proposition. Let $P \subset M_{\mathbb{R}}$ be a rational polytope of dim rank(M). Then Σ_P forms a proper fan. If the origin is in the interior of P, then

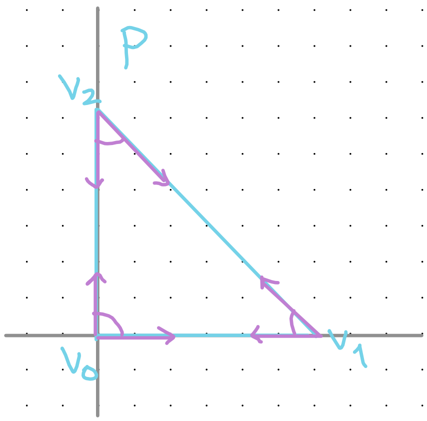
$$\Sigma_P = C_{P^\circ}$$

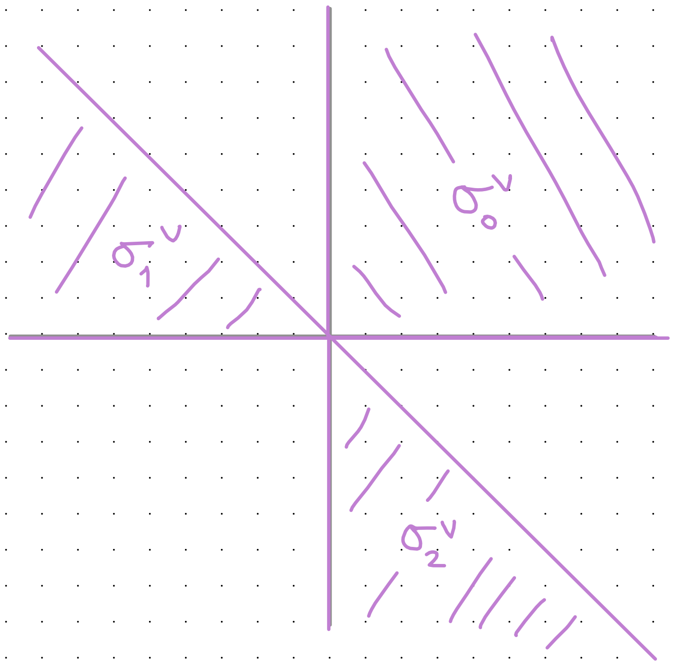
Notation. If $P \subset M_{\mathbb{R}}$ is a rational polytope, we write

$$X_P = X_{\Sigma_P}$$

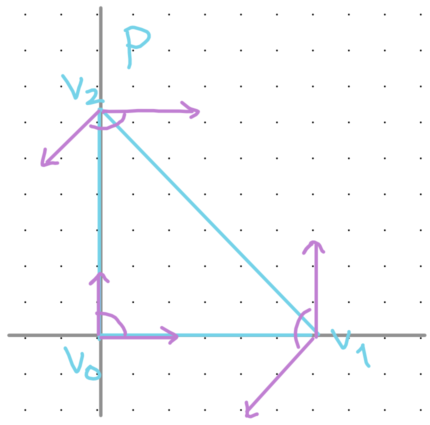
Ex. If $P \subset \mathbb{R}^3$ is the cube with vertices $(\pm 1, \pm 1, \pm 1)$, then Σ_P is the fan of cones over the faces of the octahedron with vertices $\pm e_i$. In this case, $X_P \simeq \mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$.

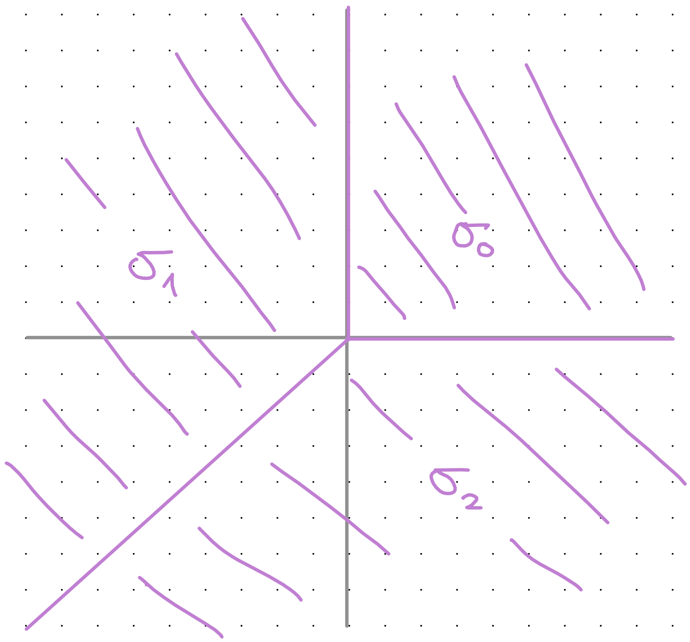
Example Consider the 2-simplex in $(\mathbb{R}^2)^\vee$ with vertices $v_0 = 0$, $v_1 = e_1^*$, $v_2 = e_2^*$. Write $\sigma_i := \sigma_{v_i}$, $\sigma_i^\vee = \sigma_{v_i}^\vee$.





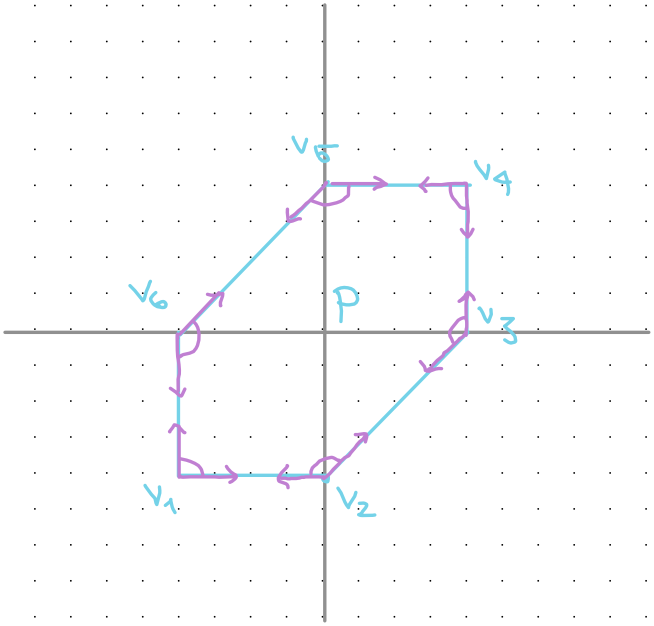
The magenta are translates of the generators for the cones $\sigma_i^\vee$. We've seen these cones before as the duals to the maximal cones for the fan we used to construct $\mathbb{P}^2$. So the fan Σ_P has maximal cones depicted below

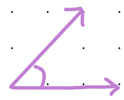




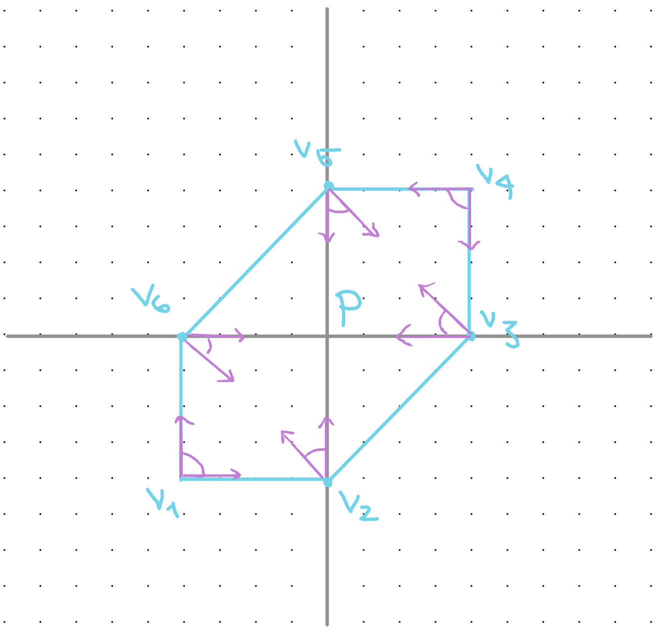
On the left, we've translated the cones σ_i to the corresponding vertices. As you can see, the cone is formed by the two vectors that are inward-pointing and normal to the edges that intersect in the vertex v_i

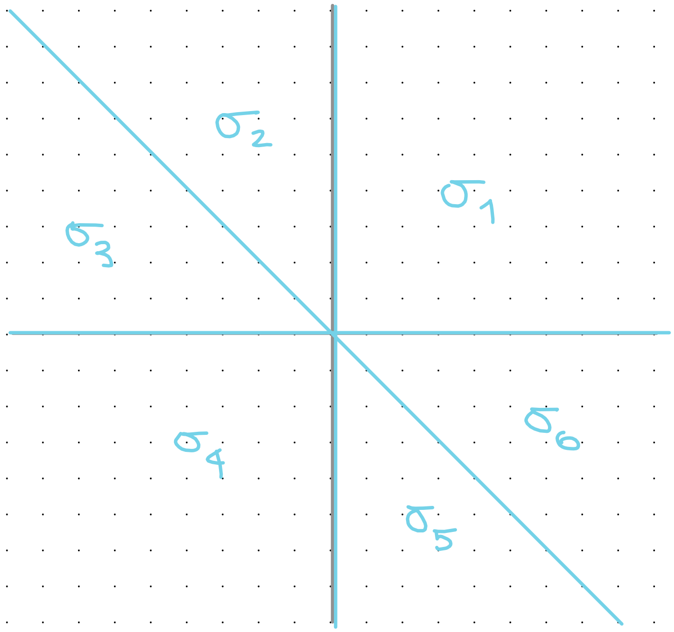
Example. Consider the polytope P below. Write $\sigma_i^\vee = \sigma_{v_i}^\vee$ and $\sigma_i = \sigma_{v_i}$.



the magenta pictures indicate translates of generators for the cones $\sigma_i^\vee$. Note that the $\sigma_i^\vee$ don't form a fan: they intersect in non faces. E.g., $\sigma_1^\vee \cap \sigma_6^\vee =$ 

Note that we've already computed the duals of (rotations of) the $\sigma_i^\vee$. Hence we see that the σ_i fit into the following picture:





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Reminder on divisors

> The next thing we want to do is to associate to a torus-equivariant divisor a piecewise linear function on $|\Sigma|$
  - This is the origin of toric mirror symmetry.

> We'll start with some reminders on divisors

Def Let X be a scheme. An **effective Cartier divisor** on X is a closed subscheme $D \subset X$ that is locally defined by a single nonzerodivisor:

(*) There are an affine open cover $\{Spec(A_i)\}_{i \in I}$ of X and nonzerodivisors $\{f_i \in A_i\}_{i \in I}$ such that

$$D \cap Spec(A_i) \cong Spec(A_i/f_i)$$

as $Spec(A_i)$-schemes

Ex The empty scheme is always an effective Cartier divisor: it is globally defined by the identity

Ex If $X \neq \emptyset$, then all of X is never an effective Cartier divisor (since 0 is a zerodivisor).

Lem Let $D \subset X$ be a closed subscheme. Then the following are equivalent:

(1) D is an effective Cartier divisor

(2) The ideal sheaf of D is a line bundle.

Def For an effective Cartier divisor $D \subset X$, we write

$$\mathcal{O}_X(-D) = \text{ideal sheaf of } D$$

$$\mathcal{O}_X(D) = \mathcal{O}_X(-D)^{\otimes -1}$$

We call $\mathcal{O}_X(D)$ the **sheaf of functions with simple poles along D**.

Remarks.

(1) Dualizing $\mathcal{O}_X(-D) \hookrightarrow \mathcal{O}_X$, we get a canonical morphism $s_D : \mathcal{O}_X \longrightarrow \mathcal{O}_X(D)$.

Explicitly, for $V \subset U$,

$$\mathcal{O}_X(D)(U) \longrightarrow \mathcal{O}_X(D)(V)$$

is injective with image local sections that locally are of the form $h = g/f$, where g and f are the the restrictions of an element $g \in \mathcal{O}_X(U)$ and a generator $f \in \mathcal{O}_X(-D)(U)$

- This justifies the terminology for $\mathcal{O}_X(D)$
- $s_D\ \mathcal{O}_X \longrightarrow \mathcal{O}_X(D)$ is the inclusion of functions with no poles along D

(2) We make the convention that $\mathcal{O}_X(D)$ has a nontrivial global section because we prefer line bundles that admit global sections.

Ntn Given effective Cartier divisors $D, D' \subset X$, we write $D+D'$ for the effective Cartier divisor corresponding to the ideal sheaf

$$\mathcal{O}_X(-D) \otimes \mathcal{O}_X(-D') \subset \mathcal{O}_X \otimes \mathcal{O}_X = \mathcal{O}_X$$

> Here's the geometric meaning

Lem. If $D, D' \subset X$ are effective Cartier divisors and the scheme-theoretic intersection $D \cap D'$ is an effective Cartier divisor on D', then $D+D'$ is the scheme-theoretic union $D \cup D'$

$$\begin{tikzcd} D \cap D' \arrow[r, hook] \arrow[d, hook] \arrow[dr, phantom, "\ulcorner", very near end] & D' \arrow[d, hook] \\ D \arrow[r, hook] & D \cup D' \end{tikzcd}$$

one can show pushouts of closed immersions along closed imm. exist